CUI WENHUI

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Education

Northwestern Polytechnical University **Xi'an, China**
Master, Software Engineering Sep 2022 – Apr 2025
Southwest Jiaotong University **Chengdu, China**
Bachelor, Network Engineering Sep 2018 – Jun 2022

- National Encouragement Scholarship (2019, 2021, for top 10%)

Experience

General Software Engineer **Jun 2025 – Present**
Huawei Technologies *Shanghai, China*

- Responsible for driving the stabilization of parallel compilation in the Rust compiler within the Rust community.
- Designed and implemented benchmarking suites for the Rust standard library and Daft to drive performance improvements on ARM; optimized core components of both projects to maximize efficiency on ARM architecture.
- Developing AI Agent-driven workflows for C/C++ to Rust conversion, empowering developers to accelerate the migration process and maintain project robustness through AI assistance.

Software Engineer Intern **Jun 2023 – Sep 2023**
Kmerit *Suzhou, China*

- Developed and enforced company policies through the creation of the "Schrute Bucks" incentive program, improving employee engagement
- Achieved record-breaking sales, earning the title of top salesperson for five consecutive years
- Supported managerial functions, including staff supervision, client relationship management, and strategic planning

Projects

Rust Benchmarking and Optimization | Rust + postgresql + Vue

- Designed and developed Rust standard library functions and Daft framework benchmarking software based on rustc-perf, and completed the web frontend development with the assistance of an AI Agent
- Conducted in-depth performance analysis using perf hotspots and assembly code inspection, identifying 5+ improvable Daft interfaces. Applied various optimization techniques including LLVM parameter tuning, manual vectorization, and reduced vtable usage, achieving a 8% performance improvement for Daft on ARM
- Deployed the benchmark software with daily scheduled execution and manual triggering via code repository webhooks

Rust Parallel Compilation Stabilization | Rust + GitHub CI

- Contributed to the Rust compiler by submitting PRs that resolved 3+ bugs related to deadlocks, data races, and non-deterministic outputs
- Added parallel compilation test support to the Rust compiler through 3+ PRs, including 20+ new test cases
- Collaborated with the Rust community to talk through design decisions and implementation details, driving the stabilization of parallel compilation and enabled by default

Web Novel Reader | Rust + Flutter + Lua

- Built a smooth and elegant UI with Flutter, Riverpod, and GoRouter, while leveraging Rust for high-performance backend logic based on Actor model
- Developed a secure, sandboxed Lua 5.4 scripting engine using mlua, providing rich built-in modules such as html (based on Rust Scraper), HTTP request abstraction, structured error handling, and support for search, catalog parsing, chapter content (with pagination), and login authentication flows
- Implemented robust local caching strategy (temporary chapter cache with auto-cleanup and permanent bookshelf) and reading progress synchronization.

Skills

Professional Skills: Sales Expertise, Leadership, Conflict Resolution, Strategic Planning, Negotiation

Specialized Talents: Karate (Black Belt), Jujitsu, Werewolf Hunting, Table Making